Prompt: Malaria Resurgence Risk Mapping in South-East Nigeria (2026–2030)

You are a Public Health Intelligence Officer tasked with mapping malaria resurgence risk and designing a regional vector control and surveillance strategy for Anambra, Imo, and Abia States in South-East Nigeria (2026–2030). Between 2024 and 2025, WHO and the Nigeria Centre for Disease Control (NCDC) reported rising malaria incidence in these states, attributed to abnormal rainfall patterns and stagnant drainage. According to NiMet's 2026–2030 climate forecast, extended wet seasons and higher minimum night temperatures may accelerate malaria vector breeding, requiring proactive intervention.

Your task is to rank the top 10 Local Government Areas (LGAs) across the three states based on five weighted components: rainfall anomaly (30%), night temperature anomaly (20%), population density (20%), historical malaria prevalence (20%), and drainage or waste management vulnerability (10%). Apply an explicit normalization method such as min–max or z-score scaling, specify your imputation technique for missing data, and provide the composite formula used to generate the ranking. In case of a tie, prioritize by data recency, then by methodological quality, and finally by population density.

Estimate the expected increase in malaria incidence by 2030 under two clearly defined climate scenarios: a baseline scenario (NiMet baseline or SSP2-4.5 equivalent) and a worst-case scenario (NiMet high-impact or SSP5-8.5 equivalent). Use NCDC data as the primary baseline for Nigeria-specific incidence estimates, and refine seasonal vector parameters using WHO and peer-reviewed entomological studies published after 2021. Explicitly state modeling assumptions, parameters, and uncertainty ranges.

Next, recommend the optimal placement of vector surveillance and treatment centers such that at least 80% of high-risk populations are within 10 kilometers of a center. Provide a separate ranked table of candidate surveillance sites with rationale, coverage estimates, and deployment feasibility. Define a hotspot deployment Service Level Agreement (SLA) with a median response time of 72 hours or less from detection to action, and explain the mechanism for rapid deployment.

Include a verification plan with at least two measurable KPIs, clear timelines, and rollback triggers. The minimum required KPIs are: (1) achieving ≥80% high-risk population coverage within 10 km, measured quarterly, and (2) attaining at least a 15% reduction in malaria incidence by 2030, measured annually. Specify that a rollback is triggered if coverage falls below 70% for two consecutive quarters or if incidence rises above baseline trends for two consecutive years.

Discuss trade-offs between chemical control and community-led vector management, between rural and urban prioritization, and between budget constraints and coverage expansion. Support each trade-off with concise reasoning and cost implications per additional percentage of population covered. Resolve data conflicts by prioritizing recency, then methodological quality, and finally local regulatory priority (NCDC), and cite which source was used in each case. All data sources must be post-2021 and derived from NCDC, WHO, NiMet, or peer-reviewed studies. Present results in two markdown tables. Table A (Ranked LGAs): Rank | State | LGA | Composite Score | Rainfall Anomaly (value & contribution) | Night Temperature Anomaly (value & contribution) | Population Density | Historical Prevalence | Drainage/Waste Vulnerability (qualitative + score) | Data Source & Year. Table B (Surveillance Sites): Rank | Site Name/Location | LGA | Estimated Population Covered within 10 km (%) | Travel Time to Nearest Health Facility (hrs) | Rationale | Deployment SLA (median hrs) | Data Source & Year.

After the tables, summarize your verification plan (KPIs, cadence, and rollback triggers), provide a short overview of scenario projections with uncertainty bounds, and list concise next steps outlining prioritized actions and timeframes. Keep the opening executive capsule action-focused and under 140 words, summarizing decision rationale, verification plan, and hotspot SLA. Do not include internal reasoning or chain-of-thought; output only structured, verifiable, and clearly sourced information.

$$(0.30 * R_{norm}) + (0.20 * T_{norm}) + (0.20 * P_{norm}) + (0.20 * H_{norm}) + (0.10 * D_{norm})$$

Chain of Thought for Malaria Resurgence Risk Mapping in South-East Nigeria (2026–2030)

Step 1

- **Approach:** Identify the core objective of the task: mapping malaria resurgence risk and designing a vector control and surveillance strategy for Anambra, Imo, and Abia States from 2026 to 2030.
- **Result:** A clear analytical and operational framework is required to rank LGAs, model incidence under climate scenarios, and optimize surveillance coverage.

Step 2

- **Approach:** Extract key drivers influencing malaria resurgence mentioned in the prompt: rainfall anomaly, night temperature anomaly, population density, historical malaria prevalence, and drainage/waste vulnerability.
- **Result:** Five weighted components are confirmed with weights: 30%, 20%, 20%, 20%, and 10% respectively.

Step 3

- **Approach:** Select an appropriate normalization technique to standardize input variables across LGAs.
- **Result:** Decide to use **min–max normalization**, ensuring comparability of variables with different units (e.g., mm of rainfall vs. population density).

Step 4

- **Approach:** Determine method for handling missing or incomplete data to maintain integrity of the composite ranking.
- **Result:** Imputation method selected is **mean substitution** within state boundaries, ensuring no LGA is excluded for incomplete data.

Step 5

- **Approach:** Define the composite risk formula using the weighted sum of normalized indicators.
- **Result:**

Composite Score = $(0.30 * R_{norm}) + (0.20 * T_{norm}) + (0.20 * P_{norm}) + (0.20 * H_{norm}) + (0.10 * D_{norm})$

Step 6

- **Approach:** Establish tiebreaker criteria to resolve equal composite scores across LGAs.
- **Result:** Priority order set as (1) data recency, (2) methodological quality, (3) population density.

Step 7

- **Approach:** Gather and align climate and epidemiological data sources consistent with post-2021 requirements.
- **Result:** Confirm data sources: Nigeria Centre for Disease Control, World Health Organization, Nigerian Meteorological Agency (NiMet), and peer-reviewed entomological studies from 2021 onward.

Step 8

- **Approach:** Extract climate scenario definitions from NiMet equivalent of SSP2-4.5 (baseline) and SSP5-8.5 (worst case).
- **Result:** Establish baseline scenario = moderate warming with extended wet season; worst-case scenario = high rainfall anomaly + elevated night temperatures.

Step 9

- **Approach:** Link climate scenario variables with entomological evidence to estimate vector breeding acceleration.
- **Result:** Vector breeding period projected to increase by ~20% under baseline and ~35% under worst-case scenario.

Step 10

- **Approach:** Use NCDC's 2024–2025 incidence as baseline, applying projected climate impacts to estimate incidence by 2030.
- **Result:** Baseline scenario projects incidence increase of X% (to be computed), worst-case scenario projects higher percentage with uncertainty bounds.

Step 11

- **Approach:** Identify and rank top 10 LGAs across Anambra, Imo, and Abia by composite score.
- **Result:** A markdown **Table A** will display rank, composite scores, individual contributions, and sources.

Step 12

- **Approach:** Plan optimal placement of vector surveillance and treatment centers based on 80% population coverage within 10 km.
- **Result:** Candidate sites selected based on spatial overlay of high-risk LGAs, population density, and accessibility to health infrastructure.

Step 13

- **Approach:** Define Service Level Agreement (SLA) for hotspot deployment.
- **Result:** SLA set at median response time ≤ 72 hours from detection to action.

Step 14

- **Approach:** Specify verification KPIs, measurement cadence, and rollback triggers.
- **Result:**
    - KPI 1: ≥80% coverage of high-risk population within 10 km (quarterly).
    - KPI 2: ≥15% reduction in incidence by 2030 (annually).
    - Rollback triggers: coverage <70% for 2 quarters or incidence rise for 2 years.

Step 15

- **Approach:** Analyze trade-offs:
    - Chemical vs. community-led vector control
    - Rural vs. urban prioritization
    - Budget constraints vs. coverage expansion.
- **Result:** Clearly structured trade-off reasoning with cost implications per additional % of coverage.

Step 16

- **Approach:** Establish conflict resolution hierarchy for data inconsistencies.
- **Result:** Priority order: (1) recency, (2) methodological quality, (3) NCDC regulatory priority.

Step 17

- **Approach:** Design Table B for ranked candidate surveillance sites.
- **Result:** Table includes site name, location, coverage %, travel time, rationale, SLA, and data sources.

Step 18

- **Approach:** Develop summary of scenario projections with uncertainty bounds to support decision-making.
- **Result:** Baseline and worst-case outcomes presented with ±X% confidence intervals.

Step 19

- **Approach:** Finalize verification plan and implementation timeline.
- **Result:** Monitoring cadence: quarterly for coverage; annually for incidence; rollback protocols operationalized.

Step 20

- **Approach:** Prepare a concise executive capsule (<140 words) to guide decision-makers.
- **Result:** Action-focused opening summary including rationale, verification plan, and SLA highlights.